

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements: *coherence* (collective phase order), *compatibility* (local descriptions agree where their domains overlap) and *self-representation* (the state carries an accurate reconstruction of itself). Coherence supplies neither of the others, and five results establish that with no cortical hypothesis: synchrony constrains content only through encoders shared on overlaps, and uniformly over pairs only inside a patch too small to span the territories at issue; the resultant does not determine the state; a unique fixed point is not a self-model; where the encoding is a region's restriction, the region must resolve the relevant family before any readout is chosen; and agreement about a change takes communication. Three conditional results replace the uniform bound by a count of the pairs that can disagree, a chain across a cover's shared sites, or a spectral gap at a frequency-locked configuration, each against a declared input coherence does not supply. Four are checkable for a machine candidate, fixing what a claim of machine unity commits its maker to measuring. Under stated hypotheses, compatible local states glue

to a unique global state that an accurate contracting reconstruction fixes uniquely; installed energy bounds a declared actuator’s mean coupling, so a coherent branch requires it to exceed $2D/\kappa$, for a κ derived here for neither cortex nor a processor. A Lean 4 development carries these results and their composition, with the eight connecting hypotheses as explicit arguments, no incomplete proofs and an audited axiom footprint. One prediction needs no absolute calibration: onset delay scales as the square root of the rate at which coupling crosses threshold, which a within-subject emergence-from-anaesthesia design can test. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling; testing it requires calibrated geometry, resolved phase dynamics and validated decoders.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. The framework’s empir-

14 ical content is the coupling-gated onset prediction and its spatial signature.
15 The identification of the latter two relations as formal correlates of conscious
16 unity and a minimal self serves as interpretive motivation: it says why the
17 mathematics might matter, but does not constitute a testable proposal with-
18 out a proposed measurement on the experiential side.

19 Unity has been approached differently elsewhere: integrated information
20 theory treats integration and exclusion as constitutive (Tononi et al., 2016),
21 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene
22 et al., 2014), and predictive-processing accounts derive both from a hierar-
23 chical generative model (Friston, 2010). Here the three requirements are
24 kept apart rather than derived from one another, so that each carries its own
25 assumptions and admits its own measurement; Section 10.3 returns to the
26 comparison.

27 Consider reaching for a cup. A visual description of its location and a
28 description of the hand’s relation to it can agree about where the cup is while
29 contributing different contents. Synchronizing the two processes would leave
30 disagreement about that shared quantity unresolved. An internal representa-
31 tion must additionally locate the perceiver within the represented situation.
32 This example motivates the requirements; the mathematical constructions
33 use spatial measures and fields in real normed vector spaces, whose identifi-
34 cation with perceptual content is a further modelling task.

35 The mathematical contribution is to specify connections between these
36 requirements and the assumptions each connection needs. A declared ac-

37 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
38 overlaps convert phase order into a bound on content disagreement. Compat-
39 ible spatial measures have a unique global extension; approximate agreement
40 controls a specified global selection. A contracting reconstruction makes that
41 extension its unique fixed point only if it accurately reconstructs the exten-
42 sion. Accuracy across a family of states imposes a further condition on the
43 information retained by the encoding.

44 The empirical contribution is to identify what observations can decide.
45 Stationary phase distributions can be shared by stronger coupling, weaker
46 diffusion and a matched common drive. Finite ramps change the observable
47 onset. A winding field is stationary at the same coupling as a uniform one and
48 has the opposite resultant, so that observable does not determine the state.
49 Coherent adaptation can lose alignment with a designated environmental
50 template, and a compatibility estimator needs diagnostics that distinguish
51 content from decoder artifacts. These findings determine the measurements
52 needed to test the proposed connections.

53 Three of the ingredients are standard: the synchronization threshold
54 $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing
55 of compatible local sections, and the Banach fixed-point theorem (Banach,
56 1922). Four things here are not. The installed-energy condition makes cou-
57 pling a purchased quantity: a declared actuator’s stored energy bounds the
58 mean coupling it supplies, so coherence on the stationary branch carries a
59 resource requirement in units a measurement can reach, and for cortex that

60 requirement reduces to a figure the mode prices must deliver. The coherence-
61 to-content bound converts phase order into a statement about descriptions
62 rather than about timing, and its patch-local form is what keeps it from
63 being vacuous in a large population. A stability criterion on the reweighted
64 coupling kernel, together with an existence condition on an amplitude layer,
65 settles when a low resultant reports weak coupling and when it reports a
66 winding field. And the composition is machine-checked together with a wit-
67 ness discharging all eight hypotheses at once, which is what makes their
68 conjunction non-empty rather than a list no object satisfies.

69 Granting those premises, collected in Table 1, the account runs as follows.
70 Extracellular contraction on awakening raises endogenous field amplitude
71 and with it the effective coupling between cortical oscillators. Once that
72 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
73 within a quarter turn the population settles onto it; other initial data leave
74 windings stationary at the same coupling. Encoders shared where cortical
75 territories overlap turn that phase order into agreement about the quantities
76 the territories describe in common, so the local descriptions glue: there is
77 one state of which each is a restriction. Part of that state encodes the state
78 itself, and a readout of that part reconstructs it accurately enough, and
79 sensitively enough across the states that matter, to make it the unique state
80 the system models itself as being. On this account the glued state is what
81 makes distributed activity one situation, and its accurate self-reconstruction
82 is what gives that situation a perspective. Every step of it is conditional.

83 Section 2 collects the results that hold whatever physical field is proposed
84 and whatever substrate carries it; a reader who rejects the cortical hypothesis
85 of Section 9 still has to accommodate them. Sections 3–5 then develop the
86 three requirements together (Fig. 1), and Section 6 reads the unconditional
87 results against a machine candidate. Section 7 composes them, stating the
88 thermodynamic constraints and all eight connecting hypotheses per branch;
89 its supporting models of feedback, learning and resource supply have distinct
90 state spaces and assumptions. Section 8 then examines what the observables
91 can establish, before Section 9 develops the cortical hypothesis and an awak-
92 ening experiment. The hypothesis is that extracellular contraction increases
93 field-mediated coupling; its direction, rate and relation to recovery require
94 independent calibration. Supplemental Material organizes the constructions
95 and controls behind the claims they support. Table 2, at the end, lists the
96 symbols and the section that defines each.

97 **2 What holds without the cortical hypothesis**

98 Most of what follows is conditional. Table 1 names eight hypotheses that
99 the composition consumes rather than derives, and the cortical identification
100 of Section 9 is a further commitment on top of them. Five results are not
101 conditional in that way. They constrain any account that reads unity off
102 phase coherence, whatever physical field is proposed to carry the coupling
103 and whatever substrate carries the field. Each is stated below where it is

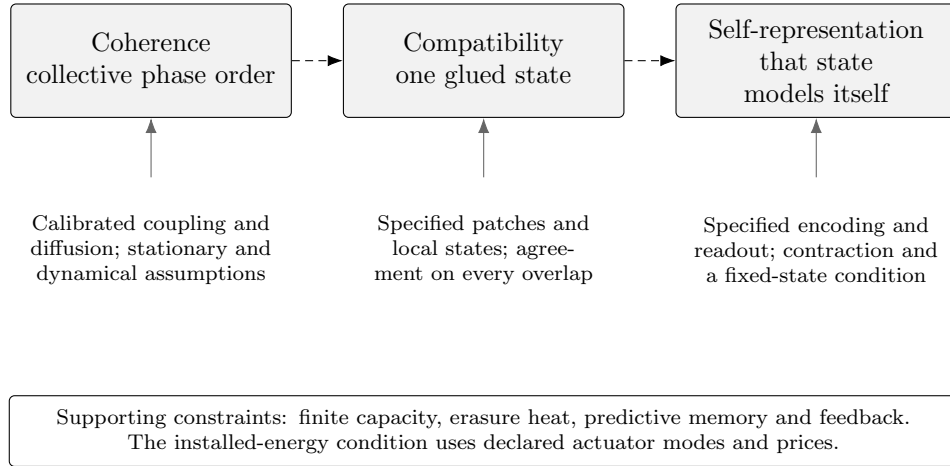


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

104 used; collected here, they are the article’s fixed points.

105 **Phase order constrains content only inside a patch too small to**
 106 **carry unity.** Assign every patch the same phase and let two local mass
 107 profiles disagree at a shared site. The profiles are chosen independently of
 108 phase, so synchrony places no constraint on their disagreement, and they
 109 have no common extension: perfect locking is consistent with local descrip-
 110 tions that do not glue (Section 4). What the positive result of Section 4.2
 111 costs is exactly this gap. A bound on content disagreement needs encoders
 112 already related on overlaps, and that relation is a supplied fact about the de-

113 scriptions rather than a consequence of the dynamics. Supplying it relocates
 114 the limit rather than lifting it. Eq. (6) bounds every pair at once and a chord
 115 never exceeds 2, so at the patch-local order of the sheet in Section 8.3 that
 116 uniform reading carries information only at 3 sites or fewer. The grain is the
 117 uniform reading's and not phase order's, and Section 4.2 states three results
 118 without it: a scale-free count of the pairs that can disagree, which names
 119 none of them; a bound chained through the sites a cover's patches share,
 120 which travels a distance linear in the hops taken; and, at a frequency-locked
 121 configuration, a bound read off the coupling graph's algebraic connectivity,
 122 which carries no population count at all. What none of the three changes is
 123 the constraint stated here: a count locates no site, and the other two con-
 124 sume a declared cover or declared hardware data, so phase order by itself
 125 still says nothing about territories too far apart to share a patch, which is
 126 where the compatibility claim is made.

127 **The order parameter does not determine the state.** A field wind-
 128 ing once across a periodic sheet has resultant $r = 0$ — the order parameter
 129 of Eq. (1) below — while every patch of it stays locked, and this is not a
 130 contrived initial condition. On a periodic array of any dimension, a winding
 131 is stationary for *any* isotropic symmetric kernel at identical frequencies, be-
 132 cause the coupling term is odd in the separation against a kernel even in it
 133 (`char_is_kuramoto_trajectory`). Section 8.3 exhibits a sheet on which uni-
 134 form and winding initial conditions reach 0.9657 and 0.0083 at one coupling,

diffusion and kernel. A low resultant therefore reports either weak coupling
or a winding field, and no measurement in r alone separates them. The state
also carries an integer the resultant does not: read around a ring, the same
winding has winding number q , and a nonzero q obstructs every global real-
valued phase lift (`winding_degree_obstructs`). The resultant is an average
over sites and moves when they are reweighted or the patches redrawn; the
degree telescopes and does not.

A unique fixed point is not a self-model. A constant readout con-
tracts, fixes whatever state it returns, and reconstructs nothing. Uniqueness
is consequently the wrong criterion, and Section 5.2 gives the right shape
of one: reconstructing M states pairwise more than 2ε apart within error ε
requires M distinguishable codes, so an alphabet of 2^b codes caps the family
at $M \leq 2^b$. This is a counting argument and holds for any encoder and
readout whatever.

**Where the encoding is a restriction, the region must resolve the
family.** Read the code as the state's own restriction to a region and
the criterion moves off the readout entirely. A boundary whose region
reads two relevant states identically cannot reconstruct them more than 2ε
apart (`not_isRestrictionResonance_of_reconstructs`), and a resonant
boundary that does reconstruct a separated family restricts injectively on
it (`restrictToAvatar_injOn_of_isRestrictionResonance`). Where the
region sits is fixed before an encoding is chosen, so no readout supplies

157 resolution the region lacks.

158 **Agreement after a change takes communication.** In a synchronous
159 network whose updates read only the site’s own state and its declared neigh-
160 bours’ messages, two configurations agreeing on a site’s causal past through
161 T rounds leave that site identical at round T . Two interventions outside
162 that past that require reports more than 2ε apart therefore cannot both be
163 reported from it within ε by the deadline (Section 5). Snapshot compatibil-
164 ity and timely compatibility are therefore different requirements, and cutting
165 the path removes the second while leaving the first intact.

166 None of these depends on the coupling being electromagnetic, on the sub-
167 strate being cortex, or on the eight hypotheses of Table 1 holding. Section 6
168 applies four of them — all but the one about the resultant, which is a state-
169 ment about a spatial observable — to a candidate that is neither biological
170 nor field-coupled.

171 3 Coherence: physical coupling and its re- 172 source condition

173 3.1 Dynamics and the stationary threshold

174 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

175 Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase dif-
176 fusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units,
177 with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; fre-
178 quency heterogeneity and diffusion are distinct parameters. The stationary
179 calculation below uses identical frequencies and this scalar coupling. Ap-
180 plying it to a spatial kernel requires an aggregation rule and a check of the
181 resulting approximation.

182 For identical frequencies in their rotating frame, the positive normalized
183 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

184 where I_n are modified Bessel functions and ψ is the collective phase. The for-
185 mal derivation uses the absence of stationary current for a periodic gradient
186 drift to obtain this density, then verifies normalization and self-consistency.

187 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above
 188 threshold, exactly one positive solution coexists with zero and increases
 189 strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

190 Figure 2 shows this stationary branch.

191 Linearization about the uniform density gives rate $K/2 - D$ for the first
 192 harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists
 193 exactly above threshold. About the coherent density, the quadratic form
 194 is strictly dissipative transverse to collective rotation, whose rate is zero.
 195 These linear estimates concern the limiting equation. Relating them to finite
 196 stochastic trajectories requires a dynamical mean-field limit and branch selec-
 197 tion; exchange symmetry alone does not establish the needed independence
 198 (Sznitman, 1991).

199 **3.2 Installed energy bounds mean coupling**

200 A spatial model replaces the finite sum by integration against a kernel. One
 201 declared implementation carries response modes with continuous profiles ϕ_i ,
 202 installation prices u_i and nonnegative occupancies $c_i(\Xi)$ in configuration Ξ :

$$K_\Xi(x, y) = \sum_i c_i(\Xi) \phi_i(x) \phi_i(y), \quad U(\Xi) = \sum_i u_i c_i(\Xi). \quad (4)$$

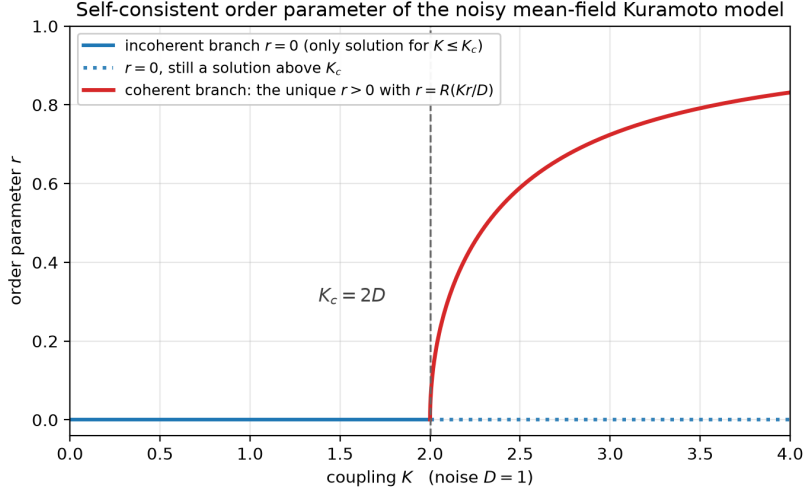


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

203 The same configuration sets the reciprocal kernel and stored energy. An
 204 executed transition changes the kernel through its changed occupancies and
 205 requires work $U(\Xi') - U(\Xi) + Q$, with Q positive into the reservoir. The
 206 installed kernel is averaged over the resulting configuration law $p(\Xi)$. Cell-
 207 pair reconstruction on refining spatial partitions converges to its product-
 208 measure integral under the specified compactness and sampling assumptions;
 209 the refinement index is independent of the number of control updates.

210 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 211 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

212 integral gives

$$\begin{aligned}
 U_{\text{inst}} &= \sum_{\Xi} p(\Xi) U(\Xi), \\
 K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_{\Xi} p(\Xi) \sum_i c_i(\Xi) m_i^2 \leq \kappa U_{\text{inst}}.
 \end{aligned}
 \tag{5}$$

213 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 214 coherent stationary branch and any growing Fourier mode about the uniform
 215 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 216 is necessary: energy above it need not produce sufficient coupling because
 217 the bound can be loose.

218 The quantity constrained is installed energy rather than dissipated heat,
 219 and the condition binds through κ rather than through the energy, since
 220 any macroscopic substrate carries installed energy far above the thermal
 221 floor. Read forwards it therefore excludes no macroscopic candidate. Read
 222 backwards it is a number. Take the region the field arithmetic takes, a
 223 sphere of one LFP decay length holding 1676 neurons, at the metabolic rates
 224 Supplemental Material collects and over one phase-diffusion time $1/D = 2$ s
 225 at $D = 0.5 \text{ rad}^2/\text{s}$: the installed energy is $1.0 \times 10^{-6} \text{ J}$, and the condition is
 226 met at every κ above $9.6 \times 10^5 \text{ J}^{-1}\text{s}^{-1}$. That figure is what an identification
 227 of cortical mode prices has to deliver, and the first identification delivering
 228 less refutes the coupling-gated account for cortex at this region and interval.
 229 Another interval rescales the figure by its ratio to this one.

230 What the condition discriminates, it discriminates through that factor.

231 The required κ scales as D^2 and inversely with the region's cell count, so at
 232 fixed mode prices the condition ranks candidate substrates by their installed
 233 energy. In a declared implementation it decides outright, the learning witness
 234 of Section 7 clearing the line after one update while its matched control stays
 235 on the exclusion side at the same conversion factor and diffusion. What is
 236 missing for cortex is a price list rather than a mechanism. A mode is a priced
 237 rank-one contribution to the kernel, cortical field generation has no canonical
 238 decomposition into such modes, and choosing the basis would choose the
 239 answer. Applying the scalar threshold to heterogeneous or signed spatial
 240 dynamics requires a justified reduction, and identifying the installed kernel
 241 with a cortical field is a separate physical commitment.

242 **4 Compatibility: from coordinated activity** 243 **to a global description**

244 **4.1 Why common timing is insufficient**

245 For overlapping patches U and V , compatibility means that restricting either
 246 local description to $U \cap V$ gives the same object. Descriptions can agree on
 247 shared quantities while contributing different information elsewhere. Com-
 248 patible sections of the specified sheaf have a unique global extension. On
 249 a finite cover, local finite spatial measures agreeing as measures on every
 250 overlap also have a unique finite-measure extension whose sheaf image is the

251 glued section. No mass-one normalization is imposed.

252 In the cup example, visual and reaching descriptions must express a
253 shared quantity, such as cup location, in common coordinates. Restriction
254 retains that quantity while dropping details particular to each description.
255 Agreement permits surface appearance and hand configuration to contribute
256 to one situation. The mathematical counterexample assigns the same phase
257 to every patch while two local mass profiles disagree on a shared site. Those
258 profiles have no common extension. Because they are chosen independently
259 of phase, synchrony places no constraint on their disagreement.

260 This distinction limits the biological interpretation. Receptive territories
261 and content domains must be related to spatial patches, and the descriptions
262 must support the required extension. Matching overlap marginals of de-
263 coded probability laws need not determine a unique global joint distribution
264 (Abramsky & Brandenburger, 2011).

265 The cover is itself a choice, and the construction constrains it in one way
266 only: synchronized patches agree where they overlap. Where they do not
267 overlap there is nothing to compare, so a cover by pairwise disjoint patches
268 satisfies that obligation for *every* assignment of local descriptions and phases
269 (`ofDisjointCover`) and the gluing theorem still returns a unique global sec-
270 tion, which is the tuple it was handed (`glue_unique_of_disjoint`). Member-
271 ship of the class then reports nothing about the system beyond the patches
272 covering it. What a gluing claim asks of a cover is therefore a non-degeneracy
273 the theorem does not assume: that distinct patches meet (`HasNonempty-`

274 Overlaps), and that what they share is a quantity fixed independently of
 275 the agreement it is being used to establish.

276 4.2 When coherence does constrain content

277 A declared relation between phase and content gives a positive result. Let
 278 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 279 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 280 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 281 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 282 sites; their equality on overlaps is a supplied relation between descriptions of
 283 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 284 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 285 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

286 Here discrepancy is the norm of the difference, and r and N are read on
 287 whichever population the bound is applied to. That choice is what decides
 288 whether the estimate says anything. A patch carries an order parameter of
 289 its own, and reading the bound there carries that patch's resultant and that
 290 patch's site count (`chord_le_of_patch_coherence`, `compatible_of_patch_`
 291 `coherence`), so a large population observed through small patches is con-
 292 strained by the small count. Read instead on the whole array, the factor N

293 makes the estimate weak, because it draws a pointwise guarantee from an
 294 average over N^2 pairs. At $r = 1$ the observations agree exactly at either
 295 scale. A shared scalar encoder is a special case. Dropping the overlap rela-
 296 tion permits disagreement even at perfect locking. A common nonexpansive
 297 linear projection preserves agreement, but agreement of a projection does
 298 not establish agreement of the full vector.

299 What Eq. (6) bounds is a chord, which never exceeds 2, so the estimate
 300 constrains anything only where $\sqrt{2} N \sqrt{1 - r^2}$ falls below that. The winding
 301 sweep of Section 8.3 fixes both ends of that range on one sheet. Read on the
 302 sweep's own patch, 625 sites at patch-local order 0.9015, the bound is $382.5 L$
 303 and constrains nothing; holding that order fixed, the count alone brings it
 304 below the maximum chord only at 3 sites or fewer. A nearest-neighbour pair
 305 of the same winding, the smallest patch there is, gives $0.1388 L$. Shrinking
 306 patches therefore thins the overlaps at the same rate as it tightens the bound,
 307 and the two ends are not reached by the same cover. At a columnar grain
 308 the estimate constrains agreement inside one locally synchronous population
 309 and is vacuous across the distant territories whose agreement unity is about.

310 That vacuity is the uniform bound's, and three results decline the step
 311 that puts N there: a sum of N^2 nonnegative terms bounded by its total,
 312 which permits all the disorder in a population to sit in the pair the conclu-
 313 sion is about. Asking instead how *many* pairs can be far apart is Markov's
 314 inequality on the same sum, and its answer carries no population count: at
 315 order r the fraction of pairs separated by a chord of c or more is at most

316 $2(1 - r^2)/c^2$ at any size of population (`chord_fraction_le`). The same sum
 317 read by rows bounds the fraction of *sites* that are far from a fixed fraction
 318 of the population (`site_fraction_le`). Read on the patch above, where the
 319 uniform bound gives $382.5 L$ and constrains nothing, this gives a fraction
 320 at most 0.37 of pairs disagreeing by more than L , and it improves like the
 321 square of what the uniform bound improves like. It is a different quantifier
 322 and not a stronger statement: a count names no pair, the exceptional set
 323 it permits is unlocated and may be exactly the sites a cover's overlaps sit
 324 on, and a guarantee about a named overlap therefore still needs the uniform
 325 bound at its own grain.

326 Distance is reached by chaining rather than by counting. Index the cover
 327 by its nerve, in which two patches are adjacent when they share a site.
 328 The chord is a metric, so a bound β on the separation inside every patch
 329 composes along a walk: a walk of k hops bounds its endpoints by $(k + 1)\beta$
 330 (`chord_le_of_patch_walk_coherence`). The count is $k + 1$ and not k because
 331 both endpoint sites pay a step inside their own patch while the shared sites
 332 are interior to the chain, and the content statement follows for any cover
 333 whose nerve has diameter k (`compatible_of_patch_nerve`). No patch need
 334 contain both sites, which is the situation of every pair of territories far enough
 335 apart, so shrinking patches and spanning territories are two ends of one chain
 336 rather than two ends of one bound. What it costs is that the accumulation is
 337 linear and nothing makes it sublinear: at the sheet's nearest-neighbour value
 338 $\beta = 0.1388 L$ the chain stays inside the maximum chord for 13 hops, which

339 is 14 patch diameters and 0.219 mm of that sheet. The territories whose
 340 agreement the compatibility claim is about are separated by centimetres,
 341 which is two orders of magnitude beyond where this chain saturates; the route
 342 reaches further only on a cover whose patches are larger or more coherent,
 343 and both are measurements. Connectivity of the nerve is a property of the
 344 cover, supplied rather than derived, and a cover whose nerve is disconnected
 345 supports no conclusion between its components.

346 The third result reads no resultant at all. Where a configuration is
 347 frequency-locked to a common rotation Ω , its phase differences are not free:
 348 each site's detuning is balanced by its coupling term, $\omega_i - \Omega = \sum_j K_{ij} \sin(\theta_i -$
 349 $\theta_j)$, which is the Kuramoto equations at a rigidly rotating solution rather
 350 than a further postulate. Pairing that balance with the configuration itself,
 351 symmetrizing, and applying Jordan's inequality termwise turns it into the
 352 coupling graph's Dirichlet form, whose algebraic connectivity λ_2 then bounds
 353 the spread about the mean phase $\bar{\theta}$ by $2\lambda_2 \|\theta - \bar{\theta}\| \leq \pi \|\omega - \Omega\|$ (`spread_le_`
 354 `of_frequency_locked`). Read at a named pair it gives a content residual
 355 of $L\pi \|\omega - \Omega\| / (\lambda_2 \sqrt{2})$ with no population count, no patch and no cover ge-
 356 ometry in it (`chord_le_of_frequency_locked`, `compatible_of_frequency_`
 357 `locked`), and it improves as the coupling graph becomes better connected,
 358 which is the dependence a resultant cannot express. Where it is informative
 359 is fixed by the same hypothesis that licenses it: confinement to a quarter
 360 turn already bounds every chord by $\sqrt{2}$ with no coupling data at all, so the
 361 spectral reading adds something exactly where $\lambda_2 > \pi \|\omega - \Omega\| / 2$. Four things

are declared and none is discharged: λ_2 is hardware data in the standing of κ ; the locked configuration is assumed, and nothing here proves one exists; the quarter-turn confinement is assumed and is not implied by locking; and the detuning enters in the population's ℓ^2 norm, so a population whose frequency spread grows with its size pays for that growth through the data. What is removed is the unconditional factor, not the physics of heterogeneity.

Structural connectivity is a declared graph and is what supplies λ_2 . Measured functional connectivity supplies it no better: it is an estimate, so it inherits the decoder failure modes of Section 8, a shared prior shrinking estimates toward agreement and regional bias manufacturing disagreement, and feeding an estimate to a hypothesis strengthens no conclusion drawn from it. Using it to choose the cover is worse than neutral, being the move of Section 4 that empties a gluing claim, in different clothes.

Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i + \eta o_i$. Content and observation discrepancies bounded by ε and δ give a next discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, reduce an initial mismatch. Approximate observations leave a nondecaying term in the error bound. Taking a uniform bound on Eq. (6) connects this update to the phase process.

383 4.3 What approximate agreement determines

384 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
385 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
386 Weights that vanish outside their patches and sum to one at each site select
387 a global profile by weighted averaging. The selection lies within ε of each
388 patch there, and any two such selections differ by at most ε , independently
389 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
390 $L\varepsilon$ across selections.

391 At zero tolerance, the selection is the unique exact extension. At every
392 positive tolerance there are families with no exact extension; selection then
393 gives an approximation whose dependence on weights is bounded. Supple-
394 mental Material gives a three-site witness attaining the bound. The mass-
395 profile construction uses the scalar case of the content dynamics and addi-
396 tionally requires nonnegativity of the initial profiles and observations. The
397 general vector result supplies no identification with the spatial-measure sheaf.

398 These constructions concern declared objects: a specified cover, contents
399 valued in a normed space, and encoders related on overlaps. They select
400 neither the neural cover nor the decoder, and identifying their contents with
401 perceptual content, or establishing that the identification bears on experi-
402 ence, is a further modelling task; Section 8 sets out what a measurement of
403 it would have to establish.

404 5 Self-representation: a state that models it- 405 self

406 5.1 The reflexive construction

407 Compatibility gives a global state. Reflexivity relates it to an encoding within
408 that state. Equip global sections with a complete metric. If the additional
409 restriction-resonance condition identifies encoding with restriction to a region
410 $\mathcal{A}$, the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (8)$$

411 The region and readout are fixed across the state space. If F contracts and
412 fixes the section s_* formed from the local descriptions, Banach’s theorem
413 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89 sup-
414 plies the contraction and $F(s_*) = s_*$; restriction resonance is separate. Su-
415 percritical coupling makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$
416 less than one for a relaxation time $\tau > 0$. Constructing a map with that
417 factor remains a modelling obligation.

418 In the cup example, the global state includes the represented scene and the
419 perceiver’s relation to it. A local encoding must retain enough information
420 for the readout to recover that modelled state; cup position alone would be
421 insufficient if hand configuration and viewpoint could not be reconstructed.

422 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-

tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (9)$$

At the stipulated rate this radius grows as $2\varepsilon/((K - 2D)\tau)$ near threshold, for fixed positive ε and τ . Exact uniqueness survives while the error guarantee weakens.

5.2 Reconstruction across a family of states

Sensitivity can be required quantitatively. Declare relevant macrostates, an encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

Thus M relevant states pairwise more than 2ε apart require at least M distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$. Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the existing self-prediction map.

This is an additional accuracy criterion on a family, with no requirement for multiple exact fixed points. A blind encoder or constant readout fails it for separated states despite having a unique fixed point.

Where the encoding is read as the state's own restriction to a region, the criterion becomes a condition on the region. Restriction resonance requires the encoder and that restriction to identify the same pairs of states, so a

441 boundary reconstructing two relevant states more than 2ε apart while its re-
 442 gion reads them identically is not resonant (`not_isRestrictionResonance_`
 443 `of_reconstructs`). Read forwards, a resonant boundary that reconstructs
 444 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
 445 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
 446 cally is fixed by where the region sits, before an encoding is chosen, so the
 447 resolution the family demands is the region's and no readout supplies it. The
 448 implication runs one way: a region that does resolve the family is not thereby
 449 shown to carry a resonant encoding, and E89 supplies no such resolution.

450 Those two demands are jointly satisfiable with nothing folded anywhere.
 451 Take the region to be the whole substrate: restriction is then the identity, so
 452 the boundary is resonant, reconstructs every declarable family exactly, and
 453 restricts injectively on all of them (`topAvatar_satisfies_obligations`). A
 454 properness condition on the region (`IsProperAvatar`) is therefore a separate
 455 requirement rather than a consequence of resonance and resolution, and what
 456 a fold into a local region is asked for. Its quantitative form replaces the
 457 alphabet by a metric. With an L -Lipschitz readout, a family reconstructed
 458 within ε whose members lie pairwise more than $d_{\min}$ apart has δ -separated
 459 codes whenever $\delta L \leq d_{\min} - 2\varepsilon$, so the family is no larger than the δ -packing
 460 number of what the region carries (`encard_le_packingNumber_restrict-`
 461 `ToAvatar`). That is the shape a continuous family of states needs, and at
 462 the whole substrate it says nothing, because the region's range is everything.
 463 The gain L is declared and derived here for no physical readout, and at

464 unbounded gain the bound says nothing.

465 **5.3 Agreement after a change takes communication**

466 Snapshot compatibility does not specify how quickly descriptions respond
467 when a shared quantity changes. In a synchronous network whose updates
468 read only the site’s own state and declared neighbours’ messages, two initial
469 configurations agreeing on a site’s causal past through T rounds produce the
470 same state there at round T . The deadline consequence is the one Section 2
471 states, and it follows from the reconstruction bound in Section 5.2: a report
472 from that site is a readout of a code that the two interventions leave identical.

473 A three-site delay line witnesses failure after one round and exact report-
474 ing after two; cutting the path prevents delivery at every round. A report
475 may still happen to be correct about one fixed world. The bound concerns
476 responsiveness across interventions.

477 Throughout this section *prediction* names a reconstruction map, and the
478 relevant family of states, the metric, the tolerance, the communication graph
479 and the deadline are all declared inputs, as are the represented variables and
480 the fidelity counted as accurate. Reading the map as temporal neural pre-
481 diction would require dynamics; reading the resulting code count as physical
482 memory capacity would require a hardware model, which a state-dependent
483 readout would have to extend.

484 6 What the conditions constrain in a digital 485 candidate

486 The results of Section 2 hold for any candidate, and for a large language
487 model (LLM) running on graphics processing units (GPUs) four of them are
488 checkable rather than merely stateable. Those four are where this framework
489 bears on claims made for such systems. It does not exclude them, and no re-
490 sult here could; what it does is fix what a claim of unity or self-representation
491 commits its maker to measuring.

492 **Codes, not reports.** Reconstruction across a family is a counting con-
493 straint before it is anything else. If two macrostates the candidate is said
494 to represent satisfy $\Phi(s) = \Phi(t)$ while $d(s, t) > 2\varepsilon$, Eq. (10) rules out re-
495 constructing both within ε , whatever the readout. A system’s reports are
496 readouts. Identical verbal reports across two conditions are therefore consis-
497 tent with identical internal codes and settle nothing about the states. The
498 converse claim—that the system represents its own condition across a de-
499 clared family—asserts that its internal code separates that family, which is a
500 statement about activations under intervention and not about text. Naming
501 the channel makes the count concrete. Where the code is the token sequence
502 a model emits—which is what its next step reads—the alphabet is the de-
503 ployed vocabulary (`tokenChannel`), and a family reconstructed through it
504 is no larger than the vocabulary size raised to the number of steps (`card`).

505 `le_card_tokens`). Over an extended exchange that number is large and the
506 bound slack. At a single step it is one vocabulary’s worth, so claiming that a
507 single emission reports the model’s own condition across a declared family as-
508 serts that the vocabulary separates it (`card_le_card_tokens_one`). A claim
509 resting on activations rather than on text has declared a different channel,
510 to which the same count applies with that channel’s alphabet. The measure-
511 ment follows from the statement: declare the family and the fidelity first,
512 then test whether the encoding region distinguishes its members.

513 **The region, not the readout.** Where the encoding is read as the state’s
514 restriction to a region of the machine, the demand falls on the region before
515 any readout exists. For a decoder-only transformer the causal mask makes
516 the network an instance of the update model of Section 5: the activation
517 at position t and layer ℓ reads positions at most t at layer $\ell - 1$ and noth-
518 ing else. A region’s causal past is therefore fixed by the architecture rather
519 than chosen by interpretation. That instance is formal, and sharper than
520 slow growth: the causal past of position t stays inside the positions at most
521 t at every depth (`mask_ball_subset_le`). The deadline bound of Section 5
522 therefore fires at every deadline at once rather than at a chosen one, its com-
523 munication graph given rather than assumed (`mask_no_guarantee`). That is
524 a statement about one forward pass; across steps a model does read its own
525 prior output, which is the channel the preceding paragraph counts. Read
526 against the coupling proposed here the asymmetry runs both ways, and nei-

527 ther way is an advantage. A kernel of full support has no neighbourhood
 528 structure for the bound to bite on: after one round the causal past is the
 529 whole substrate, so the hypothesis that fires the obstruction is unsatisfiable
 530 at any deadline but zero (`not_outside_past_of_isFullSupport`). A physi-
 531 cal field propagates at a finite speed, so treating one that way is a calibration
 532 against the phase time scale and not a theorem. What the comparison does
 533 fix is where the bound has force. Not on the attention graph, where a posi-
 534 tion reads every position at or before it and the fan-in is the context length.
 535 On the physical interconnect, fan-in is genuinely bounded: a graph whose
 536 sites each hear from at most ν others reaches at most $(T + 1)\nu^T$ sites in
 537 T rounds (`card_ball_le_mul_pow`), so a model sharded across many devices
 538 pays a number of rounds growing like $\log_\nu$ of the device count. Buying that
 539 deadline back costs wiring: reaching every site in one round requires a device
 540 hearing from all but one of the others (`le_degree_of_reaches_one`). Both
 541 ν and the device count are declared, as the communication graph and the
 542 deadline already are, and what the count bounds is the graph a guarantee
 543 requires rather than what hardware is buildable. A claim that a model re-
 544 ports on its own processing consequently carries an obligation to say which
 545 positions and layers the report's causal past contains. A report that appears
 546 to track an intervention outside it indicates a confound or a different region,
 547 not a faster channel.

548 **Agreement about what.** Unity requires the candidate’s local descriptions
549 to agree on independently identified shared quantities (Section 4). Common
550 timing, a shared residual stream and fluent output supply none of that, for
551 the reason Section 2 gives: coordination bounds content disagreement only
552 through encoders already related on overlaps. Which quantities two parts
553 of a model describe in common is a decoding question, carrying the failure
554 modes of Section 8. The cover is a choice, and choosing it to secure agreement
555 empties the claim.

556 **What is not excluded.** Digital representations can satisfy compatibility,
557 so finite hardware is compatible with the gluing construction and none of
558 the above is a no-go for machines. The coupling-gated route E56–E78 is a
559 different matter: it requires a calibrated physical coupling, its noise regime
560 and a connection from coordination to content agreement, and an attention
561 matrix or a shared clock supplies none of these identifications. Computing a
562 field model likewise leaves its realization in the computing substrate to be es-
563 tablished. Those physical obligations fall on the cortical proposal identically.
564 The capacity comparison those support comparisons lack is available, and it
565 favours neither side by construction. A continuum code space read at reso-
566 lution δ holds at least its volume divided by the volume of a resolution cell,
567 so it out-resolves an alphabet of 2^b codes exactly when that volume exceeds
568 2^b cells (`two_pow_lt_packingNumber_of_lt_measure`). Below that threshold
569 the inequality runs the other way, and at a coarse enough resolution the re-

570 gion holds a single code. What decides the comparison is the volume, the
571 resolution and the dimension, all measured quantities, and continuity decides
572 nothing on its own: a continuum code space read at a finite resolution is a fi-
573 nite alphabet whose capacity is a bit count. For the same reason this account
574 credits an analog medium with no computation beyond a Turing machine's.
575 Real-valued states carry unbounded capacity only at unbounded precision,
576 and the noise floor $D > 0$ that Eq. (1) is stated against removes it.

577 The installed-energy condition of Section 3.2 is restricted in the same
578 way. It constrains a macroscopic candidate only through κ , which is hard-
579 ware data supplied by mode profiles and prices, and Supplemental Material
580 reads the condition against a cortical metabolic budget to return the conver-
581 sion factor it would require rather than a verdict. The hardware comparisons
582 there have similarly restricted scope: fixed wiring can impose a correlation
583 disadvantage, and point-supported kernels vanish under an atomless contin-
584 uum measure, but neither comparison makes the representation conditions
585 unattainable on GPUs. A universal exclusion would need a necessary condi-
586 tion for consciousness and a proof that every relevant implementation violates
587 it, and the present composition supplies neither.

588 Whether experience requires a living substrate is the explicit subject of
589 biological naturalism (Seth, 2026a,b). On that view the properties ground-
590 ing experience are those of self-sustaining organisms, and simulating them
591 does not secure them. The conditions stated here name no biological mech-
592 anism, but computation described as such does not satisfy them either: a

593 candidate must exhibit a physically realized coupling in a measured noise
594 regime, agreement about independently identified shared quantities, and an
595 internal reconstruction meeting a declared fidelity. Whether a realization
596 of those conditions that is not alive would be accompanied by experience
597 is the point at issue, and these results do not settle it. A related position
598 locates the difference in the mode of computation rather than in life as such:
599 biological systems compute across scales that do not separate, and along-
600 side discrete operations carry continuous-valued dynamics belonging to their
601 physical medium (Milinkovic & Aru, 2026). The coupling condition stated
602 here is one measurable instance of that requirement, fixed to a physical field
603 in a declared noise regime, and a description of such a field does not satisfy
604 it.

605 The neuroscientific case against *human-like* consciousness in text-only
606 LLMs concerns embodied sensory interaction, thalamocortical organization
607 and biological regulation linking perception and action to survival (Aru et al.,
608 2023). Their absence supports skepticism, conditional on their causal rele-
609 vance to experience. Those mechanisms are not requirements of the abstract
610 gluing and reconstruction definitions, and workspace-like representations re-
611 ported in some LLMs (Gurnee et al., 2026) supply candidate organization to
612 investigate. What this framework adds to that investigation is an order for
613 the questions: declare the family of states and the fidelity, fix the encoding
614 region and read its causal past off the architecture, and only then ask what
615 the system reconstructs. A claim that survives those three steps is a claim

616 about the machine; one that does not survive them was a claim about its
617 output.

618 **7 Thermodynamics and composition**

619 The phase and representation results concern coupling, content disagreement
620 and reconstruction error. Thermodynamics constrains the physical opera-
621 tions that could implement them, and it does so from the side rather than
622 along the spine: no edge below derives coherence, compatibility or reconstruc-
623 tion from a heat bound, and the resource condition these edges terminate in
624 is necessary without being restrictive for any macroscopic candidate (Sec-
625 tion 3.2). What this half contributes is a ledger in which a resource claim
626 is stated in checkable units. Finite-register capacity obeys $H(p) \leq \log |X|$;
627 erasure heat depends on specified system–bath assumptions (Landauer, 1961;
628 Norton, 2005; Shenker, 2000). A separate boundary result concerns a field
629 reaching disconnected vacuum components (Kibble, 1976). The composition
630 connects these objects through the explicit hypotheses in Table 1.

631 **7.1 Feedback changes the information constraint**

632 For passive memory X_t driven by an environmental signal S_t , the predictive
633 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

634 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
635 et al., 2012). Zero retained information also satisfies it, so a task and a
636 learning rule require additional specification. Relating this bound to Lan-
637 dauer heat requires a common thermal scale and budget, and memory waste
638 bounded by erased entropy.

639 An acting agent changes the signal it will receive. Specify policy, world,
640 observation and memory-update channels. For their finite laws,

$$\begin{aligned} I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\ &\quad - I(X_t; S_{t+1} \mid S_t). \end{aligned} \tag{12}$$

641 The second conditional term vanishes in the passive model. With feedback,
642 the signed difference can be negative as action creates correlations with the
643 future. A causal reading requires an environmental state sufficient to describe
644 the relevant dynamics.

645 For a controlled substep, the active thermodynamic bound instead uses
646 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
647 $\Delta H(X, S) + \langle Q \rangle / (k_B T) \geq 0$ under local detailed balance, with thermal scale
648 $k_B T > 0$ and heat positive into the reservoir (Horowitz and Esposito, 2014;
649 Ito and Sagawa, 2013). Every forward path of positive probability must have
650 a positive-probability reverse for this finite identity. A missing reverse gives
651 infinite divergence. The supplement develops the support conditions and
652 composes actuation with memory updates on their shared intermediate law.

653 **7.2 Learning and repeated operation need explicit re-** 654 **sources**

655 The finite examples distinguish selecting a feasible policy, relaxing a register
656 toward an encoded reward, and learning an unknown task from observa-
657 tions. The observational learner acquires correlation with an environmental
658 parameter while its action marginal stays uniform. A continuing model exe-
659 cutes actuation, observation-driven learning and reset in sequence, preserving
660 above-chance performance at a positive work cost per cycle.

661 A phase-derived sensor makes the informational distinction concrete: an
662 informative readout and a blind control have the same phase order but dif-
663 ferent learning performance. Attaching the learner’s register to the actuator
664 of Section 3.2 ties reward, installed kernel and work to its own executed law.
665 The learning witness rises above the necessary installed-energy bound while
666 its matched control remains on the exclusion line. These finite constructions,
667 detailed in Supplemental Material, leave the evolving feedback between phase
668 dynamics, installation and sensing to be specified.

669 For that model, common energy observable $\mathcal{E}$, thermal scale $k_B T$, local
670 detailed balance and positive support at each stage give

$$k_B T [H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N \leq B + \langle \mathcal{E} \rangle_0 - \langle \mathcal{E} \rangle_N, \quad (13)$$

671 where expected cumulative work stays within the supplied allowance B at
672 every prefix. This bounds expectations. Separate constructions track a store

on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles, while continuing operation requires replenishment. Register–bath witnesses also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 3.2 requires physical relations that these separate models do not supply automatically.

7.3 What the composition establishes

The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named substep at the register’s thermal scale satisfies

$$k_B T > 0, \quad k_B T [H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

with B the register’s heat. The upper allocation follows from a ledger whose operations deliver that total heat to one reservoir and whose other operations do not increase joint entropy. Those ledger assumptions are physical inputs. Neither Landauer’s lower bound nor microscopic reversibility supplies them. The regime in which those premises are exhibited is a near-equilibrium one:

692 strictly positive forward and reverse elementary paths, local detailed balance
 693 at one thermal scale, and entropy differences and heats of order $k_B T$, the
 694 witnesses being two-bit systems with interaction energies of that size. Cortex
 695 is not in it. The installed energy of Section 3.2 is 2.4×10^{14} times kT at body
 696 temperature over one region and one phase-diffusion time, and a budget B
 697 of that size stands against an entropy term of order $k_B T$, so Eq. (14) is
 698 satisfied by any macroscopic candidate and excludes none of them. Reading
 699 the register instead at the scale where its premises are physical, such as a
 700 single channel, relocates the difficulty to E45Active, which would then have to
 701 carry one molecular step’s budget to the convergence of a continuum kernel.
 702 The active branch is an alternative route to the same node rather than a
 703 stronger one, and no result of Section 3.2 passes through it: the installed-
 704 energy condition is derived from E56 and E67.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.

Table 1 (continued)

Connection	Status	Required assumption
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness. Both are exhibited at thermal-scale heats on finite state spaces with positive reverse paths, which is not the regime a cortical register occupies.
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Convergence follows from mesh refinement rather than from the budget, which selects no regime at macroscopic scale. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.

Table 1 (continued)

Connection	Status	Required assumption
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 3.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

705 Both branches also take a thermodynamic cover carrying exact over-
706 lap compatibility and a reflexive boundary specifying encoding and readout.
707 They conclude **UnifiedSelf**: the glued section is the unique fixed point of

708 self-prediction. Compatibility and accuracy at that section enter as premises;
709 uniqueness follows. Identifying encoding with restriction needs the separate
710 resonance condition of Eq. (8).

711 Each of the eight edges is supplied rather than derived from the one above
712 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
713 pendent roots rather than a step from one to the other; both E45 discharges
714 take their convergence from refinement data and not from the thermody-
715 namic bound; E78 states that coherence relaxes the supplied cover, and the
716 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
717 coders already shared on overlaps. The inferential content of the composition
718 is its last step, where contraction and a fixed glued section give uniqueness.
719 What composing them establishes is that the premises are consistent, named
720 in one place and machine-checked together, and not that coherence follows
721 from installed energy or self-representation from coherence.

722 Joint witnesses combine a register, boundary, refining mesh and small
723 spatial cover; active witnesses add controlled thermal updates and a priced
724 installation. They establish satisfiability under their stated identifications.
725 Mesh convergence follows from refinement data, and the witnesses establish
726 no common cortical mechanism. Supplemental Material maps the claims to
727 their Lean declarations and records the counterexamples. The development
728 declares no additional axioms beyond its libraries' foundational axioms.

729 8 What the observations establish

730 The composition holds under its premises. Whether observation can estab-
731 lish those premises is a separate question, and the three requirements it
732 connects concern different objects: phase dynamics, local descriptions and
733 internal encodings. Their observables must retain these distinctions. The
734 following controls show which mechanisms remain indistinguishable, which
735 temporal approximations fail, which states one observable cannot separate,
736 and what independent content measurements must establish. Each is a finite
737 numerical regime on a declared model, with full parameters and controls in
738 Supplemental Material, and none of them calibrates a cortical quantity.

739 8.1 Stationary distributions leave the mechanism un- 740 determined

741 Equation (2) gives a concentration–coherence relation with no fitted shape
742 parameter. Estimate r from circular resultant length and a from density
743 shape: inverting R from the measured r would make agreement automatic.
744 Even independently estimated agreement identifies a statistical family shared
745 by different mechanisms.

746 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
747 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
748 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
749 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,

750 and diffusion D_0 have that same density. Their stationary distributions co-
 751 incide window by window despite different causes.

752 Resolved dynamics supply additional information. In the ideal model with
 753 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
 754 increment means constrain drift. Separating endogenous coupling from a
 755 matched common drive additionally requires a measured drive or controlled
 756 perturbation; identifying electromagnetic coupling requires evidence specific
 757 to that interaction. Supplemental Material gives the equivalence construction
 758 and its numerical check.

759 **8.2 Finite ramps change the observable onset**

760 The square-root law in Eq. (3) describes stationary solutions. A changing
 761 coupling requires time for the phase distribution to approach those solutions.
 762 In a deterministic mean-field control that crosses the threshold symmetrically,
 763 none of the scanned rates meets the declared tracking tolerance: slowing
 764 the crossing does not remove the terminal residual under that initialization.
 765 A slow control variable therefore supplies no general guarantee of branch
 766 tracking at onset. Supplemental Material gives the residual, rate range and
 767 admissible rates away from threshold.

768 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
 769 at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three
 770 consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay
 771 escape (Fig. 3); only 5/32 replicas escape within the fastest window. The

772 6 uncensored legs supply the delay scaling below; the decade-spaced ones
 773 among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted
 774 onset exponents differ from those of the exact stationary branch evaluated
 775 on the same grids. At the fastest speed the ensemble mean never reaches the
 776 fit window, so its fitted slope does not estimate stationary onset.

777 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
 778 makes the apparent exponent depend on ramp speed and fitting window.
 779 The specified deviation-based critical ramp speed is not bracketed by this
 780 sweep. With fluctuating coupling, the relevant stationary comparison also
 781 depends on rate: the mean coupling approximates order in the tested fast
 782 regime, while averaging stationary order over coupling works for the tested
 783 slow drives confined to one side of threshold. Slow crossing drives can satisfy
 784 neither approximation.

785 **The prediction.** The delay scaling is what this section supports as a pre-
 786 diction, and it needs no absolute calibration. Over the 6 uncensored legs the
 787 mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of [0.394, 0.494],
 788 which does not contain 1/2. What that shortfall measures is the escape cri-
 789 terion and not the scaling. Applied to a deterministic mean-field run seeded
 790 at the fluctuation floor, the same estimator returns 0.425 over the same legs
 791 at the same escape criterion, inside the interval on the measurement. That
 792 run has no noise to delay it, so the shortfall is the criterion's: it crosses the
 793 saturation between the floor it starts from and the level it is stopped at, and

794 tightening the criterion to $r \geq 0.05$ returns 0.490. The bias is reproduced
795 where the answer is known, which is what licenses reading the measurement
796 as the law rather than as a competing exponent.

797 Stated as a law rather than as a number: onset delay scales as the square
798 root of the rate at which coupling crosses threshold. Testing it requires
799 only that $\dot{K}$ be monotonic in a manipulated rate—not its value, nor the
800 installation factor κ , nor the field-to-coupling conversion rate γ of Eq. (15).
801 It is therefore a within-subject design varying the rate of emergence from
802 anaesthesia alone, each participant serving as their own control. Such a
803 design must avoid the hazard this sweep exhibits: a fixed absolute escape
804 criterion meets a size-dependent critical fluctuation floor, so the criterion
805 has to be referred to each recording’s own pre-threshold fluctuation scale
806 rather than to a shared number. The diffusion D is the remaining input,
807 and Section 8.1 notes that it is estimable from resolved phase traces through
808 the rate $2D$ at which quadratic variation accumulates, which is a different
809 quantity from the concentration the propofol exercise could not range over.

810 Additional controls address integration bias, population size and spatial
811 or signed coupling; the refined signed-network crossing bracket $K_{\text{eff}}/D \in$
812 $[1.958, 2.009]$ contains 2.

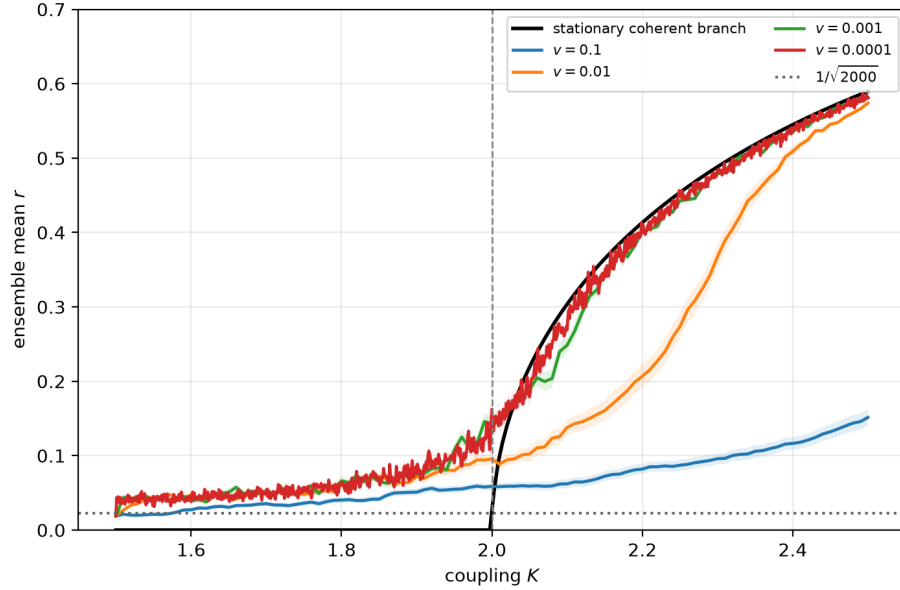


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

8.3 A winding state leaves the order parameter ambiguous

The order parameter r is indifferent to where a phase sits in space. A field winding once across a periodic sheet therefore has $r = 0$ while every patch of it stays locked. Such a field is not an artificial construction here. On a torus a winding is stationary for any isotropic symmetric kernel at identical frequencies: the coupling term is odd in the separation against a kernel even in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves

821 this for any finite periodic array, this sheet included. A winding satisfies the
822 positivity and identical-frequency hypotheses. It violates only the quarter-
823 turn initial spread that the convergence result additionally requires before
824 placing the limit at the uniform state.

825 On the sheet of Section 3's spatial control, a winding survives below
826 an interpolated decay length of 0.1871 mm and unwinds above it (Fig. 4).
827 That boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same
828 sheet at the same coupling, diffusion and kernel reaches $r = 0.9657$ from a
829 uniform start and $r = 0.0083$ from a winding one. Both are stationary, with
830 patch-local order 0.9015 and no winding defect. At fixed K the observable
831 therefore takes two values spanning nearly its whole range. Coupling does
832 not determine the resultant, and a low resultant does not by itself distinguish
833 weak coupling from a winding field.

834 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
835 trol carries, the boundary moves to 0.1421 mm. The same comparison at
836 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
837 Heterogeneity shortens the range that sustains a wave without removing it.

838 Which windings persist has a criterion behind it. Linearizing the phase
839 model about a winding returns a coupling of the same translation-invariant
840 form, each separation reweighted by the cosine of the phase the winding
841 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
842 again even in the separation, so the same characters diagonalize it and ev-
843 ery mode's rate is one of its eigenvalues. Where the winding advances by

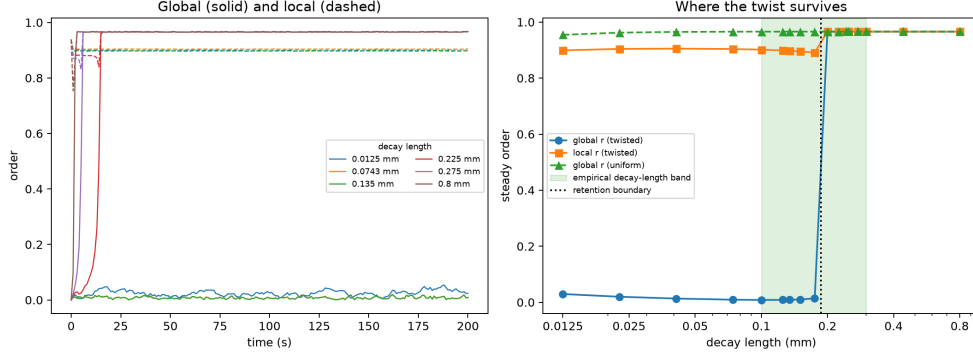


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

844 at most a quarter turn across every separation the coupling reaches, the
 845 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 846 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 847 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 848 longer separations is what carries a phase advance past a quarter turn, and
 849 that is the mechanism the decay-length sweep varies. The criterion bounds
 850 rates rather than trajectories: asymptotic stability is not proved, so the
 851 states that persist occupy a narrower range than it admits, and the bound-
 852 ary quoted above is measured and not derived from it. The same kernel is the
 853 potential's curvature, so a winding at $r = 0$ is not the global minimum the

854 convergence result places its limit at, and one carrying a growing mode is not
 855 a local minimum either: that result's quarter-turn hypothesis must exclude
 856 a winding rather than merely decline to cover it (`char_not_potential_min`).

857 An amplitude layer decides which of these states exist, and leaves what
 858 the resultant reports unchanged. Every oscillator of the phase model sits on
 859 the unit circle, so a field with an amplitude node has no representation in
 860 those dynamics. The real Stuart–Landau extension supplies one on the same
 861 coupling, and leaves intact the gradient structure the convergence results rest
 862 on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\Gamma - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its
 863 phase equation is the Kuramoto equation with each coupling weighted by a
 864 ratio of amplitudes (`amplitude_phase_is_kuramoto`). The phase model is
 865 this one with the amplitude ratios set to one, rather than a separate object.

866 What the added degree of freedom decides is which windings exist at all.
 867 On a periodic array of any dimension, this sheet included, a winding requires
 868 $\Gamma + \lambda > 0$, with Γ the layer's linear gain. Here $\lambda = \sum_d f_d(\cos \Delta\theta_d - 1)$ is
 869 the eigenvalue the kernel supplies, summed over separations, with $\Delta\theta_d$ the
 870 phase the winding advances across d (`exists_charState_iff`; on the ring
 871 $\Delta\theta_d = 2\pi qd/n$). For the nonnegative kernels used here $\lambda \leq 0$, and moving
 872 coupling mass to longer separations lowers it further. A broad enough kernel
 873 therefore removes the winding outright. The sweep above reaches that same
 874 boundary by a different route: the phase model retains every winding at
 875 every coupling, and loses this one under integration instead. The two layers
 876 bound the same regime from opposite sides. A phase recording does not

877 separate a state the amplitude layer excludes from one the dynamics fails to
878 sustain.

879 Three limits bound the layer for a reader of the cortical literature. These
880 states are windings and not spirals, and on the sheet that is a theorem rather
881 than a shortage of dimensions. Equal modulus at every site puts the phase
882 circulation at exactly zero on every plaquette, and on a periodic sheet the
883 circulations of *any* field sum to zero, so a lone defect is not a state of this
884 geometry at all. Relaxational dynamics has no propagation speed, which is
885 the quantity a cortical wave measurement leads with. And $|z|$ is a normal-
886 form radius, which a measured envelope is not; this work identifies the two
887 nowhere. Supplemental Material gives the sweep, its uniform control, the
888 amplitude layer, the standing wave it admits inside the band, the potential's
889 curvature at a winding and the defect theorems, together with the remaining
890 limits of both.

891 A vanishing resultant is the absence of global agreement and not of agree-
892 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a discrep-
893 ancy bound of $2\sqrt{2} L |\sin \Delta\theta|$, with $\Delta\theta$ the phase advanced across the patch,
894 which falls to zero as the same winding is spread over more sites (`chord_le_`
895 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
896 Encoders shared on overlaps therefore still bound content disagreement in-
897 side a patch of a state that no global observable calls coherent. The winding
898 number itself stays unconnected to content. It is, however, a property of the
899 state rather than of the cover read on it. On a ring the whole turns dropped

900 by the winding's affine local lifts are zero at every site but the wrap, where
901 the residue resets, and they sum around the loop to q ; a nonzero q obstructs
902 every global real-valued phase lift, so one state at one coupling has resultant
903 exactly zero and degree q (`winding.degree.obstructs`). Reweighting the
904 sites moves the first quantity and redrawing the loop's intermediate transi-
905 tions leaves the second where it was. The scope is the ring's: this is a loop's
906 winding and not a spiral, the transitions are those of the affine lifts rather
907 than of an unwrapped measurement, and the integer is identified with no
908 cortical observable.

909 This qualifies what a phase trace can establish. An onset criterion stated
910 in r alone registers a change of basin as readily as a change of coupling.
911 The protocol of Section 9.3 therefore needs a spatial observable beside the
912 resultant.

913 **8.4 Coherent adaptation can lose environmental align-** 914 **ment**

915 An adaptive coupling rule might supply representational structure, but its
916 objective must reward the relevant agreement. For symmetric coupling, pair-
917 wise sine terms cancel in the population mean drift. Minimizing squared drift
918 at that fixed mean therefore rewards equalizing drift across nodes. In the
919 frequency-cluster control, this favors coupling between unlike-frequency clus-
920 ters, whereas the diagnostic environmental template rewards coupling within

921 clusters.

922 The formalization proves descent of the squared-drift objective for a spec-
923 ified gradient flow at fixed phases. Its functional differs from thermodynamic
924 entropy production: at a coherent stationary density the probability current
925 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
926 rial gives the operator, its gradient and the conditions of the descent result.

927 The finite control co-evolves noisy phases and symmetric, nonnegative
928 couplings at fixed total resource. Its template is a diagnostic covariance of
929 latent signals, never a training input. Across three seeds, second-half order
930 remains between 0.9167 and 0.9222, while the within-cluster to between-
931 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5).
932 Random-update, permuted-gradient and frozen-kernel controls are reported
933 in Supplemental Material.

934 The immediate post-update objective reduction disappears when aver-
935 aged over complete update intervals. A prespecified grid of 6 rate–cadence
936 combinations contains no case meeting the 5% sustained-reduction criterion
937 while retaining coherence and losing alignment. The finding is coherent adap-
938 tation away from the specified structure, with no demonstrated sustained cost
939 reduction.

940 Kernel alignment and overlap compatibility are different observables.
941 This control addresses acquisition of its designated template under its par-
942 ticular objective. Learning models can explicitly include data in an objective
943 (Corberi et al., 2026); an account of agreement about the cup must specify

944 the content its learning rule is meant to preserve.

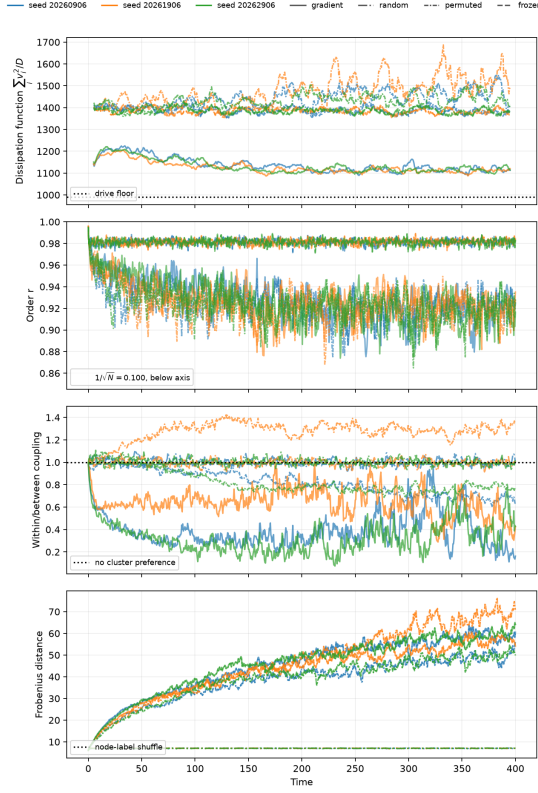


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

945 8.5 Content agreement needs independent decoders

946 A neural test needs independently identified domains, local descriptions and
947 restriction maps. Two decoders should estimate a shared content on over-
948 lapping domains before their agreement is examined. Discrepancy requires
949 a null under decoding noise and controls using deliberately compatible and
950 incompatible examples. An implementation on constructed data separates
951 these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by
952 trial; the compatible construction scores 0.476 on an interval containing one
953 half. Shared-prior shrinkage can nevertheless manufacture agreement, and
954 regional bias can manufacture disagreement. Per-patch decoding diagnostics
955 detect these failures; the compatibility statistic alone does not.

956 Phase agreement cannot replace that measurement. On constructed con-
957 tent, phase carries a share 1.000 of the content entropy when content is built
958 from it and 0.000 when it is not. A phase-derived statistic reports com-
959 patibility on phase-locked patches whose local descriptions disagree. The
960 implemented content statistic detects their incompatibility. These controls
961 validate a measurement on known constructions; a neural content model and
962 decoder remain to be established.

963 Choosing the cover also chooses a candidate subject: trivial descriptions
964 can agree while carrying little content, and a partition can conceal disagree-
965 ment. The useful choice must therefore be constrained by what neural ac-
966 tivity encodes. The exact composition assumes compatibility of its supplied
967 cover. The content mechanisms above identify additional hypotheses under

968 which agreement can arise; they leave the empirical choice of content and
969 the connection to that cover explicit.

970 **8.6 Estimator calibration on exploratory EEG**

971 The concentration range and the dependence structure decide whether the
972 concentration-coherence relation can be tested at all. Under the specified
973 log-density estimator and bias-floor criterion, discrimination at the observed
974 concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned
975 designs, rising to 31000 when sites share a mean in groups of 10. These
976 counts describe that estimator and dependence model; the intended recording
977 requires its own calibration. Supplemental Material reports the bin-count
978 controls and the full dependence-aware design.

979 That ceiling comes from exercising the estimators on public propofol-
980 sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms win-
981 dows. Bipolar and common-reference montages produce different residuals.
982 Bipolar-montage scalp EEG sits in the linear Bessel regime ($a_{\max} = 0.542$,
983 deviation from $a/2 < 1\%$). The exercise confirms the estimator is well-
984 behaved in this regime but does not test the nonlinear prediction. The
985 bipolar results are compatible with the Bessel relation and with any mono-
986 tone alternative in this regime. Temporal pooling changes the estimand, and
987 sedation recordings leave the awakening hypothesis untested.

988 An informative dataset therefore needs sufficient concentration range or
989 precision, calibrated estimators and an observation model for dependence,

source mixing and reference choice. Testing the nonlinear prediction requires intracranial recordings (ECoG, sEEG or LFP) where local synchrony reaches $a > 1$, or narrowband alpha-spindle burst analysis, at the sample sizes established above ($\geq 31,000$ pooled phase samples or ≥ 1000 independent sites). Public mouse calcium imaging provides thousands of spatially located neurons across waking, sleep and anesthesia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used here. Site count, state coverage and bandwidth are separate requirements.

9 The cortical hypothesis and an awakening test

The cortical proposal identifies a physical source of coupling and asks whether its changes help explain recovery. The tests in Section 8 determine what observations must distinguish before a phase trace can support this identification.

9.1 Physical identification and calibration

The cortical hypothesis identifies part of effective coupling with endogenous extracellular electromagnetic interactions. Fields can modulate spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models propose coordination of distributed neural information (Pinotsis & Miller,

1009 2022; McFadden, 2020). Phase organization under propofol motivates study-
1010 ing this regime (Bastos et al., 2021), but does not measure coupling.

1011 The formal field predicate requires a jointly continuous kernel, a normal-
1012 ized substrate, coupling equal to the kernel average, and no single site pro-
1013 viding more than half that strength. Toy constructions satisfy it. A cortical
1014 realization must specify the scale at which the kernel and its reduction ap-
1015 ply: calcium imaging reports different state-dependent modularity patterns
1016 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

1017 Cortical columns are the obvious candidate unit, and they answer one
1018 of the two demands the formalism makes. As the oscillator index they are
1019 a serious proposal: they are delineated by response properties and anatomy
1020 rather than by the agreement they would be used to establish (Mountcastle,
1021 1997), and a phase can be estimated from a column-sized territory at the
1022 electrode spacings used for cortical wave measurements (Townsend et al.,
1023 2015). As the content cover they are a poor one, for three separate reasons.
1024 A parcellation into non-overlapping columns is a disjoint cover, which satisfies
1025 the compatibility obligation without constraining anything (Section 4). The
1026 cover condition additionally requires the patches to exhaust the substrate,
1027 while much of association cortex has no accepted columnar parcellation. And
1028 the column’s standing as a canonical cortical unit is contested (Horton &
1029 Adams, 2005; Rakic, 2008). Offering columns for both roles at once therefore
1030 supplies the oscillator index and leaves the content cover to be identified on
1031 separate evidence, which is the bridge assumption E78 records.

1032 The spatial reduction is also testable. A row-sum aggregation at zero
1033 diagonal tracks the uniform-coupling control over the tested spatial decay
1034 range, while frequency heterogeneity displaces the operational threshold. The
1035 fit window biases both estimates relative to $2D$; agreement with the control
1036 is therefore the supported conclusion. Supplemental Material reports the
1037 reduction, its numerical error and the distinct assumptions needed for the
1038 kernel-average identification in Eq. (5).

1039 The field commitment requires physical evidence beyond a choice of math-
1040 ematical representation, and the available support comparisons do not supply
1041 it. A finite matrix spread over positive-mass cells preserves its total cou-
1042 pling and phase correlation exactly, while point-supported vanishing under
1043 an atomless measure and a disadvantage from excluding a better-correlated
1044 pair concern specified support constraints; Section 6 reads the same compar-
1045 isons against a digital candidate. Neither supplies energy prices.

1046 A spike-time shift alone supplies no coupling rate. The dimensionless
1047 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
1048 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (15)$$

1049 This rate describes how induced timing shifts accumulate into drift. Nei-
1050 ther it nor the installation factor κ follows from a heat bound. Testing the
1051 physical threshold requires independent estimates of the interaction kernel,

1052 its aggregation and diffusion in the same cortical regime.

1053 **9.2 Extracellular geometry and recovery**

1054 The proposed control variable is extracellular geometry. Extracellular vol-
 1055 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
 1056 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
 1057 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
 1058 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 1059 an effective-medium description predicts that a larger extracellular fraction
 1060 raises conductivity and reduces field amplitude.

1061 The coupling-gated hypothesis consequently predicts that extracellular
 1062 contraction on awakening raises E and, holding the other factors in Eq. (15)
 1063 fixed, raises effective coupling. This argument supplies a direction; the
 1064 geometry–coupling relation, its magnitude and its role in human recovery
 1065 require measurement.

1066 For a smooth crossing at nonzero speed, evaluating the stationary branch
 1067 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

1068 This follows from Eq. (3) at fixed D and concerns the stationary branch
 1069 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 1070 dently justified tracking approximation. The crossing controls in Section 8.2

1071 leave that approximation unsupported at onset, even though a separation of
1072 geometry and phase time scales can support it away from threshold. The
1073 awakening hypothesis therefore needs a calibrated dynamical prediction for
1074 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
1075 does not supply a recovery latency.

1076 **9.3 A protocol for the awakening experiment**

1077 The physical test has six requirements:

- 1078 **1. Coupling and diffusion in the same units.** Estimate them from re-
1079 solved phase dynamics, separating heterogeneity and observation noise.
1080 Measure the drive or perturb the system to distinguish endogenous cou-
1081 pling.
- 1082 **2. Geometry calibrated to coupling.** Validate the geometry observ-
1083 able and its relation to effective coupling, with uncertainty. The con-
1084 ductivity argument supplies only the expected direction.
- 1085 **3. Calibrated phase measurements.** Estimate r by resultant length
1086 and a by unweighted regression of log density on centered cosine.
1087 Choose site count and binning from a dependence-aware calibration
1088 over the intended concentration range. The tested designs require
1089 thousands of sites at the observed concentrations; temporal pooling
1090 changes the spatial estimand.

- 1091 **4. An independent recovery endpoint.** Measure behavioural recovery
1092 separately from phase organization.
- 1093 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-
1094 old above the finite-size floor and evaluate its latency under the mea-
1095 sured coupling trajectory, including ramp speed or fluctuations. Record
1096 a spatial observable beside the resultant, since a winding field reports
1097 a low resultant at coupling that sustains local order (Section 8.3).
- 1098 **6. Matched model comparison.** Compare coupling increase, diffusion
1099 change, common drive and alternative recovery models under the same
1100 observation and filtering assumptions.

1101 Failure of the concentration–coherence relation over an informative range
1102 constrains the density model. Failure of the calibrated geometry–coupling
1103 association constrains the physical gate. Better prediction of independent
1104 recovery by a competing mechanism counts against its proposed role. A
1105 favorable outcome still leaves the content of coordinated activity to be ex-
1106 plained.

1107 **10 Discussion**

1108 Two things have been established here, and they are of different kinds. The
1109 constraints of Section 2 hold outright: synchrony does not deliver compatible
1110 content, and where shared encoders let it bound content at all, the bound

1111 that holds uniformly over pairs does so only inside patches too small to
 1112 span the territories at issue, and distance costs either a connected cover or
 1113 a locked configuration with a declared spectral gap. The resultant does not
 1114 determine the state; a unique fixed point is not a reconstruction; a region that
 1115 cannot resolve a family cannot encode it; and agreement about a change takes
 1116 communication. Any account that reads unity off phase coherence has to
 1117 meet them, this one included. The account set out in Section 1 is conditional
 1118 at every step by contrast: each connection in it is a hypothesis supplied
 1119 rather than a result derived, and Table 1 names all eight. Its four non-
 1120 standard ingredients, named in Section 1, are the installed-energy condition,
 1121 the patch-local form of the content bound, the criterion separating a winding
 1122 field from weak coupling, and the machine-checked composition with its joint
 1123 witness. What follows takes the conditions in turn, as commitments the
 1124 proposal incurs, as measurements that would bear on them, and as questions
 1125 the framework leaves open.

1126 **10.1 Unity and a minimal self as interpretive motiva-** 1127 **tion**

1128 We identify the state glued from compatible local descriptions as a formal
 1129 correlate of conscious unity, and its accurate internal reconstruction as a
 1130 correlate of a minimal self. This identification is the framework's explicit
 1131 interpretive motivation: it explains why the mathematics might matter. The

1132 motivations are that different contents belong to one experienced situation
1133 and that the situation is experienced from a perspective. The framework’s
1134 empirical content, however, is the coupling-gated onset prediction and its spa-
1135 tial signature. The conditional results identify relations a realization would
1136 need: sufficient installed resources for the specified phase model, agreement
1137 about shared content, and an encoding that supports reconstruction at the
1138 required fidelity. Discharging E12–E89 for cortex would make the composi-
1139 tion a statement about cortex: that a cortical field above threshold relaxes
1140 to a cover whose compatible local descriptions glue, and that the glued state
1141 is the unique fixed point of a reconstruction map realized there. It would
1142 leave the two identifications above exactly where they are, because neither is
1143 among the eight and neither follows from their conjunction; they are the mo-
1144 tivation this subsection names, and without an independent measurement for
1145 experience, they do not form a testable proposal. Section 10.2 lists the em-
1146 pirical observations that would count against the framework’s mathematical
1147 claims.

1148 Compatibility allows different descriptions to contribute different infor-
1149 mation. It also permits uncertainty and conflicting judgments: apparent cup
1150 position and judged cup position can be distinct quantities in a consistent
1151 description of their discrepancy. Which quantities are shared must be de-
1152 termined independently. Choosing them solely to secure agreement would
1153 deprive the proposal of empirical force.

1154 The minimal-self proposal concerns the perceiver’s position and condition

1155 within the situation, such as the cup’s location relative to a body that can
1156 reach it. The encoding could be implicit and need carry no autobiograph-
1157 ical narrative. Its proposed necessity is philosophical: even an unreflective
1158 episode has a perspective. Gluing by itself requires no self-encoding, and re-
1159 constructability alone supplies no functional role. Altering the encoded hand
1160 or viewpoint while holding the external scene fixed should change estimates
1161 of the perceiver’s relation to the cup as the readout predicts. Whether that
1162 represented perspective is experienced remains a further question.

1163 **10.2 What a test of the full proposal must establish**

1164 Both uniqueness results concern specified objects. They choose neither the
1165 cover nor the encoding, and do not select between candidate subjects. In-
1166 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
1167 here selection requires independent evidence about neural organization. A
1168 constant readout can contract and fix a predetermined state while discard-
1169 ing its input. The family criterion in Section 5.2 tests the sensitivity that
1170 this construction lacks, provided the relevant states and fidelity are declared
1171 independently.

1172 A test must separately measure coupling relative to diffusion, content
1173 compatibility and self-reconstruction, alongside an independent endpoint for
1174 experience. Meeting all three without a unified episode would challenge suffi-
1175 ciency. A unified episode with incompatible descriptions would challenge the
1176 unity identification; one lacking the specified self-encoding would challenge

1177 the reflexive requirement. Awakening, dreaming and dissociative states offer
1178 possible settings, but absence of report cannot by itself define absence of
1179 experience.

1180 The next physical requirement is a calibrated geometry–coupling relation
1181 and resolved phase dynamics. The installation bound contributes a nec-
1182 essary resource condition where its hardware representation applies; cortical
1183 mode prices and field-mediated coupling require measurement. For the repre-
1184 sentational connection, an informative experiment asks whether a controlled
1185 change in coordination alters independently decoded content agreement, with
1186 controls for sensory information and decoder quality. The constructed-data
1187 estimator establishes feasibility for specified contents. Neural local descrip-
1188 tions, shared overlap encoders and the self-encoding have yet to be identi-
1189 fied. The population-size dependence of the phase-to-content bound limits
1190 its quantitative use in the form Section 4.2 reads off the sheet, and each
1191 way around it asks the measurement for something further: a located excep-
1192 tional set, before a count of disagreeing pairs bears on a named overlap; a
1193 cover whose nerve is connected and whose patch diameters are measured; or
1194 a frequency-locked population over a coupling graph whose connectivity is
1195 measured structurally rather than estimated from the activity the bound is
1196 about.

1197 The separation results explain why these measurements must be inde-
1198 pendent. A stationary distribution leaves several recovery mechanisms indis-
1199 tinguishable. Equal phase order can accompany incompatible descriptions

1200 or different task information. The tested plasticity rule preserves coherence
1201 while losing its designated template. Each finding concerns the specified
1202 model and observable; their role is to delimit the evidence needed for the
1203 proposed connections.

1204 **10.3 Explanatory scope**

1205 Integrated-information, global-workspace and predictive-processing accounts
1206 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
1207 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
1208 candidate physical interaction and conditional relations between local content
1209 and self-encoding. Gluing and reconstruction permit multiple substrates;
1210 the electromagnetic commitment enters through physical calibration. That
1211 permission is wider than the unconditional results of Section 2 alone. The
1212 conditional chain is substrate-neutral in the same way: its field condition
1213 asks for a jointly continuous kernel on a normalized substrate, a coupling
1214 equal to that kernel’s double average, a noise that is the field’s own, and
1215 no single site carrying more than half the coupling—and none of these dis-
1216 tinguishes an electromagnetic kernel from any other continuous mean-field
1217 kernel. The site condition is the only one with physical flavour, and it holds
1218 vacuously wherever the substrate has no atoms (`no_site_dominates`), which
1219 is the continuum case the electromagnetic hypothesis is about. The chain
1220 therefore transfers to extracellular ion diffusion, to gap junctions modelled
1221 as a density, or to a mechanical or optical medium, modulo the calibration of

1222 Section 9.1. What carries the electromagnetic identification is that calibra-
1223 tion together with the extracellular-geometry argument of Section 9, which is
1224 the one place a field equation enters. The formalization checks implications
1225 under their hypotheses, and its witnesses establish realizability on declared
1226 models. Two things stand outside that result and are not supplied by it: the
1227 eight hypotheses are discharged for toy components rather than for cortex,
1228 and the identification of the glued state with unity and of its reconstruction
1229 with a minimal self is a further commitment that the eight do not contain.

1230 The explanatory question is whether the specified relations occur in neural
1231 activity and matter for experience. In the cup example, this means establish-
1232 ing how coordination helps maintain one represented situation and how the
1233 perceiver’s place within it contributes to the episode. The proposal identi-
1234 fies candidate correlates and tests of that identification. Why those relations
1235 would be accompanied by experience remains unexplained.

1236 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	§2, Supp.	Phase diffusion in inverse time; phase spread $D \leq \pi/2$

Table 2 (continued)

Symbol	Defined	Meaning
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
μ	Eq. (4)	Substrate measure, normalized to one
Ξ, ϕ_i, u_i, c_i	Eq. (4)	Actuator configuration; mode profile, installation price and occupancy of mode i
U_{inst}, κ	Eq. (5)	Installed energy and the mode-data conversion factor bounding $K_{\text{mf}} \leq \kappa U_{\text{inst}}$
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (10)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (8)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time
T	§3.2, §5	Temperature in the thermal scale $k_B T$; number of communication rounds to a deadline
ν	Section 6	Fan-in of a communication graph: the most sites any one site hears from
$k_B T, B$	Eq. (14)	Thermal scale, read as one symbol, and heat budget of a register
$\gamma, \mathcal{T}, f$	Eq. (15)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction
λ	§8.3, §9	Eigenvalue the coupling kernel supplies at a winding; extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover's nerve, in hops

Table 2 (continued)

Symbol	Defined	Meaning
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (10)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (8)	Encoding region and the reconstruction map $F(s) = \text{readout}(s \mathcal{A})$
$\mathcal{E}$	Eq. (13)	Energy observable shared by the stages of a continuing protocol
z_i, Γ, A_{ij}	Section 8.3	Complex amplitude, linear gain and coupling matrix of the amplitude layer
$\Delta\theta_d$	Section 8.3	Phase a winding advances across separation d
E	Eq. (15)	Field amplitude, in the cortical identification
U	§3.2	Stored energy $U(\Xi)$ of a configuration
s, s_*	§4–5	A global section and the one the reconstruction fixes

Data and code availability

The Lean development, simulation scripts, tests and saved numerical summaries are maintained in the GitHub repository `bobaseb/coupling_gated_cortical_coherence`. Publication simulation values are generated from saved summaries without rerunning production sweeps. The exploratory EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-processing and selection are described in Supplemental Material. The repository contains the reproducible analysis and does not redistribute the raw recordings.

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1247 This research received no external funding. The author declares no compet-
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1249 **Use of AI tools**

1250 AI assistance to this work was substantial and is not separable by task. Sev-
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1253 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
1254 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
1255 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
1256 accessed with the Hermes CLI. Between them they contributed to the Lean
1257 development, the simulation and analysis code, and the drafting and revision
1258 of both documents. Because the tools were used in combination and in alter-
1259 nation, no section, proof, figure or result is attributable to any one of them,
1260 and none is claimed to be. The author is responsible for all content, including
1261 everything a tool produced. Machine checking establishes the stated Lean
1262 results under their hypotheses; it does not validate biological interpretations
1263 or replace scientific review of generated code and prose.

References

- Pinotsis, D. A., & Miller, E. K. (2022). Beyond dimension reduction: Stable electric fields emerge from and allow representational drift. *NeuroImage*, 253, 119058.
- Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7), 450-461.
- Baars, B. J. (1997). In the Theater of Consciousness: The Workspace of the Mind. *Oxford University Press*.
- Dehaene, S., Charles, L., King, J. R., & Marti, S. (2014). Toward a computational theory of conscious processing. *Current Opinion in Neurobiology*, 25, 76-84.
- Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature Reviews Neuroscience*, 11(2), 127-138.
- Milinkovic, B., & Aru, J. (2026). On biological and artificial consciousness: A case for biological computationalism. *Neuroscience & Biobehavioral Reviews*, 181, 106524. doi:10.1016/j.neubiorev.2025.106524.
- Aru, J., Larkum, M. E., & Shine, J. M. (2023). The feasibility of artificial consciousness through the lens of neuroscience. *Trends in Neurosciences*, 46(12), 1008–1017. doi:10.1016/j.tins.2023.09.009.

- 1284 Seth, A. K. (2026). Conscious artificial intelligence and biological naturalism.
1285 *Behavioral and Brain Sciences*, 49, e315. doi:10.1017/S0140525X25000032.
- 1286 Seth, A. K. (2026). The stuff matters: Consciousness, compu-
1287 tation, and biology. *Behavioral and Brain Sciences*, 49, e366.
1288 doi:10.1017/S0140525X26106906.
- 1289 Gurnee, W., et al. (2026). Verbalizable Representations Form a
1290 Global Workspace in Language Models. *Transformer Circuits Thread*.
1291 arXiv:2607.15495.
- 1292 Aguilera, M., Millidge, B., Tschantz, A., & Buckley, C. L. (2022). How par-
1293 ticular is the physics of the free energy principle?. *Physics of Life Reviews*,
1294 40, 24-50.
- 1295 Biehl, M., Pollock, F. A., & Kanai, R. (2021). A technical critique of some
1296 parts of the free energy principle. *Entropy*, 23(3), 293.
- 1297 Bruineberg, J., Dolega, K., Dewhurst, J., & Baltieri, M. (2022). The Em-
1298 peror’s New Markov Blankets. *Behavioral and Brain Sciences*, 45, e183.
- 1299 Pinotsis, D. A., & Miller, E. K. (2023). In vivo ephaptic coupling allows
1300 memory network formation. *Cerebral Cortex*, 33(17), 9877-9895.
- 1301 Pinotsis, D. A., Fridman, G., & Miller, E. K. (2023). Cytoelectric coupling:
1302 Electric fields sculpt neural activity and ”tune” the brain’s infrastructure.
1303 *Progress in Neurobiology*, 226, 102465.

- 1304 McFadden, J. (2002). The conscious electromagnetic information (cemi) field
1305 theory: the hard problem made easy?. *Journal of Consciousness Studies*,
1306 9(8), 45-60.
- 1307 Pockett, S. (2012). The electromagnetic field theory of consciousness: a
1308 testable hypothesis about the characteristics of conscious as opposed to
1309 non-conscious fields. *Journal of Consciousness Studies*, 19(11-12), 191-223.
- 1310 McFadden, J. (2020). Integrating information in the brain's EM field: the
1311 cemi field theory of consciousness. *Neuroscience of Consciousness*, 2020(1),
1312 niaa016.
- 1313 Fröhlich, F., & McCormick, D. A. (2010). Endogenous electric fields may
1314 guide neocortical network activity. *Neuron*, 67(1), 129-143.
- 1315 Anastassiou, C. A., & Koch, C. (2015). Ephaptic coupling to endogenous
1316 electric field activity: why bother?. *Current Opinion in Neurobiology*, 31,
1317 95-103.
- 1318 Pockett, S. (2002). Difficulties with the electromagnetic field theory of con-
1319 sciousness. *Journal of Consciousness Studies*, 9(4), 51-56.
- 1320 Vöröslakos, M., et al. (2018). Direct effects of transcranial electric stimulation
1321 on brain circuits in rats and humans. *Nature Communications*, 9, 483.
- 1322 Verkhratsky, A., & Nedergaard, M. (2018). Physiology of Astroglia. *Physio-*
1323 *logical Reviews*, 98(1), 239-389.

- 1324 Mountcastle, V. B. (1997). The columnar organization of the neocortex.
1325 *Brain*, 120(4), 701-722. doi:10.1093/brain/120.4.701.
- 1326 Horton, J. C., & Adams, D. L. (2005). The cortical column: a structure
1327 without a function. *Philosophical Transactions of the Royal Society B*,
1328 360(1456), 837-862. doi:10.1098/rstb.2005.1623.
- 1329 Rakic, P. (2008). Confusing cortical columns. *Proceedings of the National*
1330 *Academy of Sciences*, 105(34), 12099-12100. doi:10.1073/pnas.0807271105.
- 1331 Bastos, A. M., Donoghue, J. A., Brincat, S. L., Mahnke, M., Yanar, J., Cor-
1332 rea, J., ... & Miller, E. K. (2021). Neural effects of propofol-induced uncon-
1333 sciousness and its reversal using thalamic stimulation. *eLife*, 10, e60824.
- 1334 Bajwa, I. J., Nilsen, A. S., Skukies, R., Aamodt, A., Ernst, G., Storm, J.
1335 F., & Juel, B. E. (2025). A repeated awakening study exploring the capac-
1336 ity of complexity measures to capture dreaming during propofol sedation.
1337 *Scientific Reports*, 15, 32746.
- 1338 Xie, L., Kang, H., Xu, Q., Chen, M. J., Liao, Y., Thiyagarajan, M., ... &
1339 Nedergaard, M. (2013). Sleep drives metabolite clearance from the adult
1340 brain. *Science*, 342(6156), 373-377.
- 1341 Syková, E., & Nicholson, C. (2008). Diffusion in brain extracellular space.
1342 *Physiological Reviews*, 88(4), 1277-1340.
- 1343 Barbour, B. (2017). Analysis of claims that the brain extracellular impedance
1344 is high and non-resistive. *Biophysical Journal*, 113(7), 1636-1638.

1345 Norton, J. D. (2005). Eaters of the lotus: Landauer's principle and the return
1346 of Maxwell's demon. *Studies in History and Philosophy of Science Part B:*
1347 *Studies in History and Philosophy of Modern Physics*, 36(2), 375-411.

1348 Landauer, R. (1961). Irreversibility and heat generation in the computing
1349 process. *IBM Journal of Research and Development*, 5(3), 183-191.

1350 Kibble, T. W. B. (1976). Topology of cosmic domains and strings. *Journal*
1351 *of Physics A: Mathematical and General*, 9(8), 1387-1398.

1352 Sakaguchi, H. (1988). Cooperative phenomena in coupled oscillator systems
1353 under external fields. *Progress of Theoretical Physics*, 79(1), 39-46.

1354 Still, S., Sivak, D. A., Bell, A. J., & Crooks, G. E. (2012). Thermodynamics
1355 of prediction. *Physical Review Letters*, 109(12), 120604.

1356 Kawai, R., Parrondo, J. M. R., & Van den Broeck, C. (2007). Dissipation:
1357 The phase-space perspective. *Physical Review Letters*, 98(8), 080602.

1358 Horowitz, J. M., & Esposito, M. (2014). Thermodynamics with continuous
1359 information flow. *Physical Review X*, 4, 031015. doi:10.1103/PhysRevX.4.
1360 031015.

1361 Ito, S., & Sagawa, T. (2013). Information thermodynamics on
1362 causal networks. *Physical Review Letters*, 111, 180603. doi:10.1103/
1363 PhysRevLett.111.180603.

- 1364 Corberi, F., dello Russo, S., Messuti, G., Scarpetta, S., & Smaldone, L.
1365 (2026). Thermodynamic learning. *arXiv preprint*. arXiv:2609.04732.
- 1366 Banach, S. (1922). Sur les opérations dans les ensembles abstraits et leur ap-
1367 plication aux équations intégrales. *Fundamenta Mathematicae*, 3(1), 133-
1368 181.
- 1369 Shenker, O. R. (2000). Logic and entropy. *PhilSci-Archive preprint*. philsci-
1370 archive.pitt.edu/115.
- 1371 Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Boyd, A. B., Berta, M., Jensen,
1372 H. J., Gastpar, M., & Carroll, S. M. (2026). Four faces of information in
1373 natural systems. *Nature Reviews Physics*. doi:10.1038/s42254-026-00971-4.
- 1374 Sznitman, A.-S. (1991). Topics in propagation of chaos. In *École d'Été de*
1375 *Probabilités de Saint-Flour XIX—1989*, Lecture Notes in Mathematics,
1376 vol. 1464, pp. 165–251. Springer. doi:10.1007/BFb0085169.
- 1377 Berglund, N., & Gentz, B. (2002). Pathwise description of dynamic pitchfork
1378 bifurcations with additive noise. *Probability Theory and Related Fields*,
1379 122, 341–388. doi:10.1007/s004400100174.
- 1380 Abramsky, S., & Brandenburger, A. (2011). The sheaf-theoretic structure
1381 of non-locality and contextuality. *New Journal of Physics*, 13, 113036.
1382 doi:10.1088/1367-2630/13/11/113036.
- 1383 Townsend, R. G., Solomon, S. S., Chen, S. C., Pietersen, A. N. J., Martin,
1384 P. R., Solomon, S. G., & Gong, P. (2015). Emergence of complex wave

1385 patterns in primate cerebral cortex. *The Journal of Neuroscience*, 35(11),
1386 4657-4662. doi:10.1523/JNEUROSCI.4509-14.2015.

1387 Xu, Y., Long, X., Feng, J., & Gong, P. (2023). Interacting spiral wave pat-
1388 terns underlie complex brain dynamics and are related to cognitive pro-
1389 cessing. *Nature Human Behaviour*, 7(7), 1196-1215. doi:10.1038/s41562-
1390 023-01626-5.

1391 Nicoletti, G., Di Terlizzi, I., & Busiello, D. M. (2026). Balancing information
1392 and dissipation with partially observed fluctuating signals. *Physical Review*
1393 *Letters*, 137(12), 127101. doi:10.1103/8gx2-rbns.

1394 Kiyooka, D., Oomoto, I., Kitazono, J., Saito, Y., Kobayashi, M., Matsub-
1395 ara, C., Kobayashi, K., Murayama, M., & Oizumi, M. (2026). Single-
1396 cell resolution functional networks during unconsciousness are segre-
1397 gated into spatially intermixed modules. *Cell Reports*, 45(2), 116902.
1398 doi:10.1016/j.celrep.2025.116902.

1399 Oomoto, I., Kiyooka, D., Oizumi, M., & Murayama, M. (2026). Multi-area
1400 single-cell calcium imaging dataset of the mouse cortex across wakefulness,
1401 sleep, and anesthesia. *Scientific Data*. doi:10.1038/s41597-026-08202-2.

1402 Attwell, D., & Laughlin, S. B. (2001). An energy budget for signaling in the
1403 grey matter of the brain. *Journal of Cerebral Blood Flow & Metabolism*,
1404 21(10), 1133-1145. doi:10.1097/00004647-200110000-00001.

- 1405 Harris, J. J., Jolivet, R., & Attwell, D. (2012). Synaptic energy use and
1406 supply. *Neuron*, 75(5), 762-777. doi:10.1016/j.neuron.2012.08.019.
- 1407 Nicholls, D. G., & Ferguson, S. J. (2013). *Bioenergetics 4*. Academic Press.